MACHINE LEARNING ALGORITHMS

BY SUMANA

Types of ml Algo:

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

1.Supervised Learning:

- Regression
- Decision Tree
- Random Forest
- KNN
- Logistic Regression
- 1.Regression
- 1.1.Linear Regression
- 1.2.Non-Linear Regression
- 2.Classification

2. Unsupervised learning

examples :

- apriori algorithm
- K-means

3.Reinforcement Learning:

Example :
Markov Decision Process

LIST OF COMMON MACHINE LEARNING ALGORITHMS

- Linear Regression {{% used to estimate real values (cost of houses, number of calls, total sales etc.) %}}
- Logistic Regression / logit regression. : {{% used to estimate discrete values (Binary values like 0/1, yes/no, true/false) / Predicts Probability %}}
- Decision Tree (type of supervised learning algo , for classification problems)
- SVM {{%**Support Vector Machine**%}}(classification method) (when we have two features like Height and Hair length of an individual)
- Naïve Bayes (classification technique based on Bayes' theorem)
- kNN **((k- Nearest Neighbors))**(used in both classification and regression problems)
- K-Means (unsupervised algorithm, solves clustering problem)
- Random Forest {{% trademark term for an ensemble/collection of decision trees. %}}
- Dimensionality Reduction Algorithms
- Gradient Boosting algorithms
  - GBM
  - XGBoost
  - LightGBM
  - CatBoost

Different type of clustering algo :

- Affinity propagation
- Agglomerative clustering
- Birch
- Dbscan
- K-means
- Knn
- Mini-batch k-means
- Mean shift
- Optics
- Spectral clustering
- Mixture of gaussians
- Principal component analysis
- Singular value decomposition
- Independent component analysis

- Distribution based methods
- Centroid based methods
- Connectivity based methods
- Density models
- Subspace clustering
- Fuzzy c-means (FCM) algorithm
- Expectation-maximisation (EM) algorithm
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- Hierarchical clustering algorithms.

- Top five algo:
- K-means clustering
- Mean-shift clustering
- Density-based spatial clustering of applications with noise (DBSCAN)
- Expectation–maximization (EM) clustering using gaussian mixture models (GMM)
- Agglomerative hierarchical clustering

Types:

- Exclusive(partitioning)
- Agglomerative
- Overlapping
- Probabilistic

Decision Tree Algorithms

- Classification and Regression Tree (CART)
- Iterative Dichotomiser 3 (ID3)
- C4.5 and C5.0 (different versions of a powerful approach)
- Chi-squared Automatic Interaction Detection (CHAID)
- Decision Stump
- M5
- Conditional Decision Trees

Regularization Algorithms

- Ridge Regression
- Least Absolute Shrinkage and Selection Operator (LASSO)
- Elastic Net
- Least-Angle Regression (LARS)

Bayesian Algorithms

- Naive Bayes
- Gaussian Naive Bayes
- Multinomial Naive Bayes
- Averaged One-Dependence Estimators (AODE)
- Bayesian Belief Network (BBN)
- Bayesian Network (BN)

Clustering Algorithms

- k-Means
- k-Medians
- Expectation Maximisation (EM)
- Hierarchical Clustering

Deep Learning Algorithms

- Convolutional Neural Network (CNN)
- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory Networks (LSTMs)
- Stacked Auto-Encoders
- Deep Boltzmann Machine (DBM)
- Deep Belief Networks (DBN)

Ensemble Algorithms

- Boosting
- Bootstrapped Aggregation (Bagging)
- AdaBoost
- Weighted Average (Blending)
- Stacked Generalization (Stacking)
- Gradient Boosting Machines (GBM)
- Gradient Boosted Regression Trees (GBRT)
- Random Forest

Instance-based Algorithms:

- k-Nearest Neighbor (kNN)
- Learning Vector Quantization (LVQ)
- Self-Organizing Map (SOM)
- Locally Weighted Learning (LWL)
- Support Vector Machines (SVM)

Artificial Neural Network Algorithms

- Perceptron
- Multilayer Perceptrons (MLP)
- Back-Propagation
- Stochastic Gradient Descent
- Hopfield Network
- Radial Basis Function Network (RBFN)

Other Machine Learning Algorithms

- Feature selection algorithms
- Algorithm accuracy evaluation
- Performance measures
- Optimization algorithms
- Computational intelligence (evolutionary algorithms, etc.)
- Computer Vision (CV)
- Natural Language Processing (NLP)
- Recommender Systems
- Reinforcement Learning
- Graphical Models
- Warnsdorff's algorithm
- Naive Algorithm
- Backtracking Algo

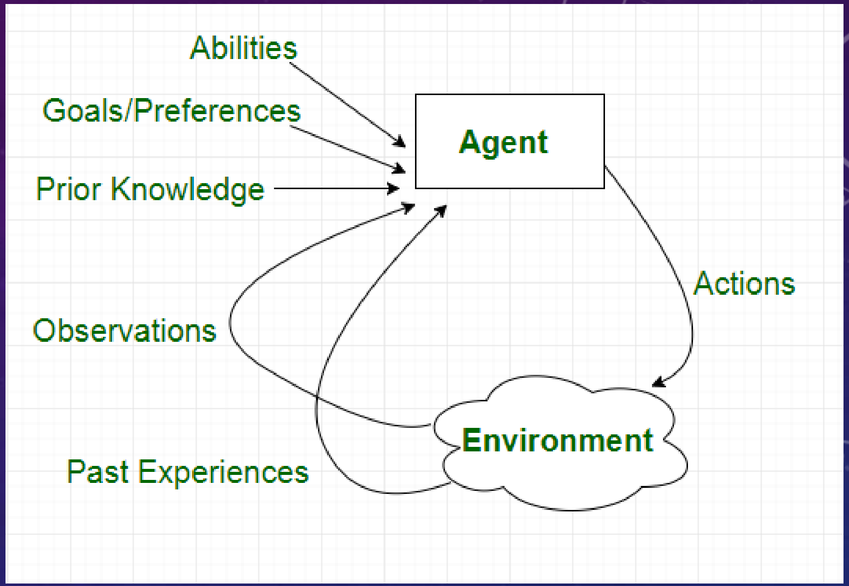
Dimensionality Reduction Algorithms

- Principal Component Analysis (PCA)
- Principal Component Regression (PCR)
- Partial Least Squares Regression (PLSR)
- Sammon Mapping
- Multidimensional Scaling (MDS)
- Projection Pursuit
- Linear Discriminant Analysis (LDA)
- Mixture Discriminant Analysis (MDA)
- Quadratic Discriminant Analysis (QDA)
- Flexible Discriminant Analysis (FDA)

Association Rule Learning Algorithms

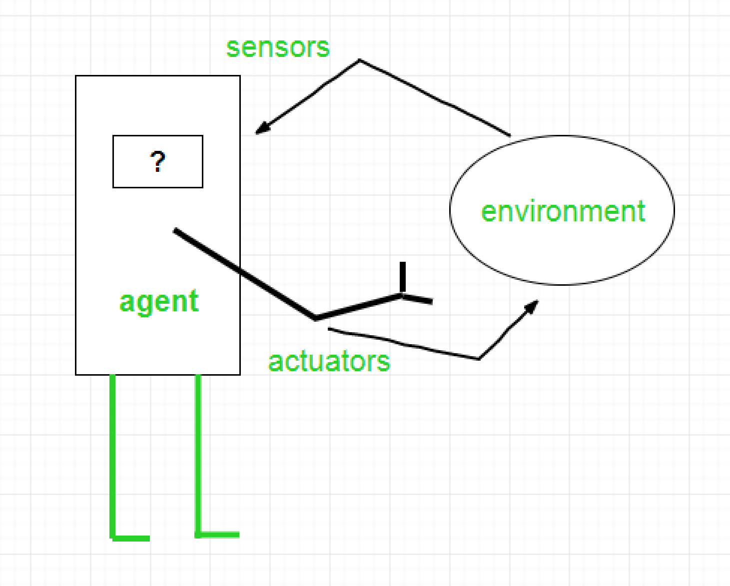
- Apriori algorithm
- Eclat algorithm

AGENTS IN ARTIFICIAL INTELLIGENCE



Types of Agents

1. Simple Reflex Agents
2. Model-Based Reflex Agents
3. Goal-Based Agents
4. Utility-Based Agents
5. Learning Agent



Examples of Agent:

1.**Software agent** : has Keystrokes, file contents-> received network packages -> act as sensors ->displays on the screen(files, sent network packets acting as actuators)

2. **Human agent** : has eyes, ears, and other organs ,act as **sensors** and hands, legs, mouth, and other body parts acting as **actuators.**

3. **Robotic agent :** has Cameras and infrared range finders which act as **sensors** and various motors acting as **actuators.**

THANK YOU